Tarea drug discovery

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Tarea relacionada a la aula de Chemberta

Los estudiantes deberan desarrollar un notebook en el cual utilicen los dados trabajados en clase (Ache_activity_IC50.csv) para crear un modelo de predicción para búsqueda de nuevos inhibidores del target mencionado.

Los estudiantes deben hacer cualquier tipo de pre-procesamiento de los datos y escoger un algoritmo (RF, SVM, etc). Puede ser escogido cualquier tipo de descriptores moleculares (propiedades moleculares, fingerprints, chemberta).

Pueden realizar clasificación o regresión. La precisión del modelo no será considerada para la evaluación, lo mas importante será el esquema de trabajo.

```
!pip install rdkit
```

```
Collecting rdkit
  Downloading rdkit-2025.3.3-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (4
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packag
Downloading rdkit-2025.3.3-cp311-cp311-manylinux_2_28_x86_64.whl (34.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 34.9/34.9 MB 11.1 MB/s eta 0:00:00
Installing collected packages: rdkit
Successfully installed rdkit-2025.3.3
```

```
import rdkit
```

```
import pandas as pd
import numpy as np
from rdkit import Chem
from rdkit.Chem import Descriptors
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
#from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.svm import SVR
from sklearn import svm
from sklearn.naive_bayes import GaussianNB
import matplotlib.pyplot as plt
from rdkit import Chem
from rdkit.Chem import AllChem
from rdkit.ML.Descriptors import MoleculeDescriptors
from rdkit import DataStructs
from rdkit.Chem import rdMolDescriptors
from rdkit.Chem.Descriptors import MolWt
from rdkit.Chem.Descriptors import MolLogP
from rdkit.Chem.Descriptors import NumHAcceptors
from rdkit.Chem.Descriptors import NumHDonors
from rdkit.Chem import Lipinski
from rdkit.Chem import QED
from rdkit.Chem import MolSurf
from IPython.display import SVG # to use Scalar Vector Graphics (SVG) not bi
from rdkit.Chem import Draw # to draw molecules
from rdkit.Chem.Draw import IPythonConsole # to draw inline in iPython
from rdkit.Chem import rdDepictor  # to generate 2D depictions of molecules
from rdkit.Chem.Draw import rdMolDraw2D # to draw 2D molecules using vectors
import seaborn as sns
```

Preprocesamiento

```
df=pd.read_csv('Ache_activity_IC50.csv', sep=";")
df.head()
```

	Molecule ChEMBL ID	Molecule Name	Molecule Max Phase	Molecular Weight	#RO5 Violations	AlogP	Compound Key	
0	CHEMBL345905	NaN	NaN	243.35	0.0	2.77	1B	
1	CHEMBL263322	NaN	NaN	271.40	0.0	3.55	10	
2	CHEMBL338492	NaN	NaN	208.28	0.0	2.12	35	
3	CHEMBL93936	NaN	NaN	405.54	0.0	4.51	1j	c1(
4	CHEMBL440185	NaN	NaN	236.27	0.0	2.09	8 (table I)	

5 rows × 48 columns

```
df.info() # contamos con 9415 entries
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9415 entries, 0 to 9414
Data columns (total 48 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Molecule ChEMBL ID          9415 non-null   object
 1   Molecule Name               1898 non-null   object
 2   Molecule Max Phase          1505 non-null   float64
 3   Molecular Weight            9403 non-null   float64
 4   #RO5 Violations             9302 non-null   float64
 5   AlogP                       9302 non-null   float64
 6   Compound Key                9415 non-null   object
 7   Smiles                      9379 non-null   object
 8   Standard Type               9415 non-null   object
 9   Standard Relation           8130 non-null   object
 10  Standard Value              8130 non-null   float64
 11  Standard Units              8151 non-null   object
 12  pChEMBL Value               6717 non-null   float64
 13  Data Validity Comment       717 non-null    object
 14  Comment                     1340 non-null   object
 15  Uo Units                    8143 non-null   object
```

```
 16  Ligand Efficiency BEI       6635 non-null   float64
 17  Ligand Efficiency LE        6595 non-null   float64
 18  Ligand Efficiency LLE       6595 non-null   float64
 19  Ligand Efficiency SEI       6575 non-null   float64
 20  Potential Duplicate         9415 non-null   int64
 21  Assay ChEMBL ID             9415 non-null   object
 22  Assay Description           9415 non-null   object
 23  Assay Type                  9415 non-null   object
 24  BAO Format ID               9415 non-null   object
 25  BAO Label                   9415 non-null   object
 26  Assay Organism              6900 non-null   object
 27  Assay Tissue ChEMBL ID      29 non-null     object
 28  Assay Tissue Name           86 non-null     object
 29  Assay Cell Type             1424 non-null   object
 30  Assay Subcellular Fraction  3 non-null      object
 31  Assay Parameters            4 non-null      object
 32  Assay Variant Accession     0 non-null      float64
 33  Assay Variant Mutation      0 non-null      float64
 34  Target ChEMBL ID            9415 non-null   object
 35  Target Name                 9415 non-null   object
 36  Target Organism             9415 non-null   object
 37  Target Type                 9415 non-null   object
 38  Document ChEMBL ID          9415 non-null   object
 39  Source ID                   9415 non-null   int64
 40  Source Description          9415 non-null   object
 41  Document Journal            8471 non-null   object
 42  Document Year               8544 non-null   float64
 43  Cell ChEMBL ID              1017 non-null   object
 44  Properties                  1270 non-null   object
 45  Action Type                 1522 non-null   object
 46  Standard Text Value         0 non-null      float64
 47  Value                       8130 non-null   float64
dtypes: float64(15), int64(2), object(31)
memory usage: 3.4+ MB
```

```
df['Standard Value']
```

	Standard Value
0	7.70
1	1.50
2	5333.35
3	0.80
4	13.18
...	...
9410	1900.00
9411	2600.00
9412	11300.00
9413	7943.28
9414	NaN

9415 rows × 1 columns

dtype: float64

```
df['pChEMBL Value']
```

	pChEMBL Value
0	8.11
1	8.82
2	5.27
3	9.10
4	7.88
...	...
9410	5.72
9411	5.58
9412	4.95
9413	5.10
9414	NaN

9415 rows × 1 columns

dtype: float64

```
# Quitamos todos los valores nulos
df = df.dropna(subset=['Standard Value', 'pChEMBL Value'])

# Aseguramos que todos sean numeric data
df['Standard Value'] = pd.to_numeric(df['Standard Value'], errors='coerce')
df['pChEMBL Value'] = pd.to_numeric(df['pChEMBL Value'], errors='coerce')

# Volvemos a eliminar filas con valores no numericos
df = df.dropna(subset=['Standard Value', 'pChEMBL Value'])
df.info() # contamos con 6717 entries
```

```
 3   Molecular Weight            6717 non-null   float64
 4   #RO5 Violations             6677 non-null   float64
 5   AlogP                       6677 non-null   float64
 6   Compound Key                6717 non-null   object
 7   Smiles                      6716 non-null   object
 8   Standard Type               6717 non-null   object
 9   Standard Relation           6717 non-null   object
 10  Standard Value              6717 non-null   float64
```

```
 11  Standard Units              6717 non-null   object
 12  pChEMBL Value               6717 non-null   float64
 13  Data Validity Comment       0 non-null      object
 14  Comment                     62 non-null     object
 15  Uo Units                    6717 non-null   object
 16  Ligand Efficiency BEI       6635 non-null   float64
 17  Ligand Efficiency LE        6595 non-null   float64
 18  Ligand Efficiency LLE       6595 non-null   float64
 19  Ligand Efficiency SEI       6575 non-null   float64
 20  Potential Duplicate         6717 non-null   int64
 21  Assay ChEMBL ID             6717 non-null   object
 22  Assay Description           6717 non-null   object
 23  Assay Type                  6717 non-null   object
 24  BAO Format ID               6717 non-null   object
 25  BAO Label                   6717 non-null   object
 26  Assay Organism              5316 non-null   object
 27  Assay Tissue ChEMBL ID      27 non-null     object
 28  Assay Tissue Name           61 non-null     object
 29  Assay Cell Type             497 non-null    object
 30  Assay Subcellular Fraction  2 non-null      object
 31  Assay Parameters            2 non-null      object
 32  Assay Variant Accession     0 non-null      float64
 33  Assay Variant Mutation      0 non-null      float64
 34  Target ChEMBL ID            6717 non-null   object
 35  Target Name                 6717 non-null   object
 36  Target Organism             6717 non-null   object
 37  Target Type                 6717 non-null   object
 38  Document ChEMBL ID          6717 non-null   object
 39  Source ID                   6717 non-null   int64
 40  Source Description          6717 non-null   object
 41  Document Journal            6620 non-null   object
 42  Document Year               6687 non-null   float64
 43  Cell ChEMBL ID              139 non-null    object
 44  Properties                  904 non-null    object
 45  Action Type                 1396 non-null   object
 46  Standard Text Value         0 non-null      float64
 47  Value                       6717 non-null   float64
dtypes: float64(15), int64(2), object(31)
memory usage: 2.5+ MB
<ipython-input-10-881242300>:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/st
  df['Standard Value'] = pd.to_numeric(df['Standard Value'], errors='coerce')
<ipython-input-10-881242300>:6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/st
```

```
df['pChEMBL Value'] = pd.to_numeric(df['pChEMBL Value'], errors='coerce')
```

```
# Checamos que  no haya valores (NaN) en Smiles
df["Smiles"].isna().any()

# Filtramos
resultado = df.dropna(subset=['Smiles'])
resultado.info() # bajamos a 6716 entries
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 6716 entries, 0 to 9413
Data columns (total 48 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Molecule ChEMBL ID          6716 non-null   object
 1   Molecule Name               901 non-null    object
 2   Molecule Max Phase          680 non-null    float64
 3   Molecular Weight            6716 non-null   float64
 4   #RO5 Violations             6677 non-null   float64
 5   AlogP                       6677 non-null   float64
 6   Compound Key                6716 non-null   object
 7   Smiles                      6716 non-null   object
 8   Standard Type               6716 non-null   object
 9   Standard Relation           6716 non-null   object
 10  Standard Value              6716 non-null   float64
 11  Standard Units              6716 non-null   object
 12  pChEMBL Value               6716 non-null   float64
 13  Data Validity Comment       0 non-null      object
 14  Comment                     62 non-null     object
 15  Uo Units                    6716 non-null   object
 16  Ligand Efficiency BEI       6634 non-null   float64
 17  Ligand Efficiency LE        6595 non-null   float64
 18  Ligand Efficiency LLE       6595 non-null   float64
 19  Ligand Efficiency SEI       6575 non-null   float64
 20  Potential Duplicate         6716 non-null   int64
 21  Assay ChEMBL ID             6716 non-null   object
 22  Assay Description           6716 non-null   object
 23  Assay Type                  6716 non-null   object
 24  BAO Format ID               6716 non-null   object
 25  BAO Label                   6716 non-null   object
 26  Assay Organism              5315 non-null   object
 27  Assay Tissue ChEMBL ID      27 non-null     object
 28  Assay Tissue Name           61 non-null     object
 29  Assay Cell Type             497 non-null    object
 30  Assay Subcellular Fraction  2 non-null      object
 31  Assay Parameters            2 non-null      object
 32  Assay Variant Accession     0 non-null      float64
 33  Assay Variant Mutation      0 non-null      float64
 34  Target ChEMBL ID            6716 non-null   object
```

```
 35  Target Name                 6716 non-null   object
 36  Target Organism             6716 non-null   object
 37  Target Type                 6716 non-null   object
 38  Document ChEMBL ID          6716 non-null   object
 39  Source ID                   6716 non-null   int64
 40  Source Description          6716 non-null   object
 41  Document Journal            6619 non-null   object
 42  Document Year               6686 non-null   float64
 43  Cell ChEMBL ID              139 non-null    object
 44  Properties                  904 non-null    object
 45  Action Type                 1396 non-null   object
 46  Standard Text Value         0 non-null      float64
 47  Value                       6716 non-null   float64
dtypes: float64(15), int64(2), object(31)
memory usage: 2.5+ MB
```

```
# Buscamos duplicados en Smiles
duplicatas = resultado[resultado.duplicated(subset=['Smiles'], keep=False)]
len(duplicatas) # 1811 duplicados
```

```
1811
```

```
# Removemos duplicados manteniendo la primera ocurrencia
resultado = resultado.drop_duplicates(subset=['Smiles'])
resultado.info() #5383 entries
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5383 entries, 0 to 9412
Data columns (total 48 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Molecule ChEMBL ID          5383 non-null   object
 1   Molecule Name               216 non-null    object
 2   Molecule Max Phase          95 non-null     float64
 3   Molecular Weight            5383 non-null   float64
 4   #RO5 Violations             5345 non-null   float64
 5   AlogP                       5345 non-null   float64
 6   Compound Key                5383 non-null   object
 7   Smiles                      5383 non-null   object
 8   Standard Type               5383 non-null   object
 9   Standard Relation           5383 non-null   object
 10  Standard Value              5383 non-null   float64
 11  Standard Units              5383 non-null   object
 12  pChEMBL Value               5383 non-null   float64
 13  Data Validity Comment       0 non-null      object
 14  Comment                     59 non-null     object
```

```
 15  Uo Units                    5383 non-null   object
 16  Ligand Efficiency BEI       5316 non-null   float64
 17  Ligand Efficiency LE        5278 non-null   float64
 18  Ligand Efficiency LLE       5278 non-null   float64
 19  Ligand Efficiency SEI       5265 non-null   float64
 20  Potential Duplicate         5383 non-null   int64
 21  Assay ChEMBL ID             5383 non-null   object
 22  Assay Description           5383 non-null   object
 23  Assay Type                  5383 non-null   object
 24  BAO Format ID               5383 non-null   object
 25  BAO Label                   5383 non-null   object
 26  Assay Organism              4310 non-null   object
 27  Assay Tissue ChEMBL ID      9 non-null      object
 28  Assay Tissue Name           40 non-null     object
 29  Assay Cell Type             440 non-null    object
 30  Assay Subcellular Fraction  0 non-null      object
 31  Assay Parameters            2 non-null      object
 32  Assay Variant Accession     0 non-null      float64
 33  Assay Variant Mutation      0 non-null      float64
 34  Target ChEMBL ID            5383 non-null   object
 35  Target Name                 5383 non-null   object
 36  Target Organism             5383 non-null   object
 37  Target Type                 5383 non-null   object
 38  Document ChEMBL ID          5383 non-null   object
 39  Source ID                   5383 non-null   int64
 40  Source Description          5383 non-null   object
 41  Document Journal            5294 non-null   object
 42  Document Year               5358 non-null   float64
 43  Cell ChEMBL ID              118 non-null    object
 44  Properties                  785 non-null    object
 45  Action Type                 1197 non-null   object
 46  Standard Text Value         0 non-null      float64
 47  Value                       5383 non-null   float64
dtypes: float64(15), int64(2), object(31)
memory usage: 2.0+ MB
```

```
# Confirmamos duplicados en Smiles
duplicatas = resultado[resultado.duplicated(subset=['Smiles'], keep=False)]
len(duplicatas) # 0 duplicados
```

```
0
```

Algoritmo

```
# Trabajamos con las 5383 entries

resultado.loc[resultado['Standard Value'] > 501, 'Activity'] = 0 #inactives
resultado.loc[resultado['Standard Value'] <= 500, 'Activity'] = 1
```

```
resultado['Activity'].value_counts()
```

Activity	count
0.0	3165
1.0	2218

dtype: int64

propiedades moleculares

```
# Usando lo visto en clase para estandarizar
def standardize_mol(mol, verbose=False):
    """Standardize the RDKit molecule, select its parent molecule, uncharge i
    then enumerate all the tautomers.
    If verbose is true, an explanation of the steps and structures of the mol
    as it is standardized will be output."""
    # Follows the steps from:
    #  https://github.com/greglandrum/RSC_OpenScience_Standardization_202104/
    # as described **excellently** (by Greg Landrum) in
    # https://www.youtube.com/watch?v=eWTApNX8dJQ -- thanks JP!

    from rdkit.Chem.MolStandardize import rdMolStandardize
    # removeHs, disconnect metal atoms, normalize the molecule, reionize the
    clean_mol = rdMolStandardize.Cleanup(mol)
    if verbose:
        #print('Remove Hs, disconnect metal atoms, normalize the molecule, re
        draw_mol_with_SVG(clean_mol)

    # if many fragments, get the "parent" (the actual mol we are interested i
    parent_clean_mol = rdMolStandardize.FragmentParent(clean_mol)
    if verbose:
        #print('Select the "parent" fragment:')
```

```
        draw_mol_with_SVG(parent_clean_mol)

    # try to neutralize molecule
    uncharger = rdMolStandardize.Uncharger() # annoying, but necessary as no
    uncharged_parent_clean_mol = uncharger.uncharge(parent_clean_mol)
    if verbose:
        #print('Neutralize the molecule:')
        draw_mol_with_SVG(uncharged_parent_clean_mol)

    # Note: no attempt is made at reionization at this step
    # nor ionization at some pH (RDKit has no pKa caculator);
    # the main aim to to represent all molecules from different sources
    # in a (single) standard way, for use in ML, catalogues, etc.
    te = rdMolStandardize.TautomerEnumerator() # idem
    taut_uncharged_parent_clean_mol = te.Canonicalize(uncharged_parent_clean_
    if verbose:
        #print('Enumerate tautomers:')
        draw_mol_with_SVG(taut_uncharged_parent_clean_mol)
    assert taut_uncharged_parent_clean_mol != None

    if verbose: print("smile_stand:", Chem.MolToSmiles(taut_uncharged_parent_
    return taut_uncharged_parent_clean_mol


def standardize_smiles(smiles, verbose=False):
  """Standardize the SMILES string, select its parent molecule, uncharge it,
    then enumerate all the tautomers."""
  if verbose: print(smiles)
  std_mol = standardize_mol(Chem.MolFromSmiles(smiles), verbose=verbose)
  return Chem.MolToSmiles(std_mol)


def draw_mol_with_SVG(mol, molSize=(450,150)):
    """Use SVG to draw an RDKit molecule, mol."""
    mc = Chem.Mol(mol.ToBinary())
    if not mc.GetNumConformers():
        rdDepictor.Compute2DCoords(mc) # Compute 2D coordinates
    # Initialize the drawer with the size
    drawer = rdMolDraw2D.MolDraw2DSVG(molSize[0],molSize[1])
    drawer.DrawMolecule(mc) # Draw the molcule
    drawer.FinishDrawing()
    svg = drawer.GetDrawingText() # Get the SVG string
    display(SVG(svg.replace('svg:',''))) # Fix the SVG string and display
```

```
# Nuevo indice
resultado = resultado.reset_index(drop=True)
smiles_ache = []

for i in range(len(resultado)):
    smiles_ache.append(resultado["Smiles"][i])


# prueba
smiles_ache[24]
```

```
'Nc1c2c(nc3ccccc13)CCCC2'
```

```
# Estandarizamos todo
smiles_ache_stand = []
for i in range(len(smiles_ache)):
    #standardize_smiles(smiles_ache[i])
    smiles_ache_stand.append(standardize_smiles(smiles_ache[i]))
```

```
[17:49:54] Running Normalizer
[17:49:54] Running LargestFragmentChooser
[17:49:54] Running Uncharger
[17:49:54] Initializing MetalDisconnector
[17:49:54] Running MetalDisconnector
[17:49:54] Initializing Normalizer
[17:49:54] Running Normalizer
[17:49:54] Initializing MetalDisconnector
[17:49:54] Running MetalDisconnector
[17:49:54] Initializing Normalizer
[17:49:54] Running Normalizer
[17:49:54] Running LargestFragmentChooser
[17:49:54] Running Uncharger
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Running LargestFragmentChooser
[17:49:55] Running Uncharger
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
```

```
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Running LargestFragmentChooser
[17:49:55] Fragment: COc1c(C(=O)CCC2CCN(CC3CCCCC3)CC2)cc(Cl)c2c1ccn2S(=O)(=O)c
[17:49:55] New largest fragment: COc1c(C(=O)CCC2CCN(CC3CCCCC3)CC2)cc(Cl)c2c1cc
[17:49:55] Fragment: O=C(O)/C=C/C(=O)O
[17:49:55] Running Uncharger
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Running LargestFragmentChooser
[17:49:55] Running Uncharger
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Initializing MetalDisconnector
[17:49:55] Running MetalDisconnector
[17:49:55] Initializing Normalizer
[17:49:55] Running Normalizer
[17:49:55] Running LargestFragmentChooser
[17:49:55] Fragment: Cl
[17:49:55] New largest fragment: Cl (2)
[17:49:55] Fragment: Cl
[17:49:55] New largest fragment: Cl (2)
[17:49:55] Fragment: N[C@@H](Cc1c[nH]c2ccccc12)C(=O)NCCCCCCCCNc1c2c(nc3ccccc13
```

```
print(len(smiles_ache), len(smiles_ache_stand))
```

```
5383 5383
```

```
SMILES_STAND_ACHE = pd.DataFrame(smiles_ache_stand)
SMILES_STAND_ACHE.to_csv('smiles_stand_ache_no_duplicates.csv', sep=';')
```

```
descritors = {'QED':[],'MolWt':[],'HeavyAtomCount':[], 'NumRotatableBonds':[]
              'NumHAcceptors':[],'NumHDonors':[]}
```

```
for i in range(len(smiles_ache_stand)):
    m = Chem.MolFromSmiles(smiles_ache_stand[i])
    descritors['QED'].append(QED.default(m))
    descritors['MolWt'].append(MolWt(m))
    descritors['HeavyAtomCount'].append(Lipinski.HeavyAtomCount(m))
    descritors['NumRotatableBonds'].append(Lipinski.NumRotatableBonds(m))
    descritors['TPSA'].append(MolSurf.TPSA(m))
    descritors['NumHAcceptors'].append(Lipinski.NumHAcceptors(m))
    descritors['NumHDonors'].append(Lipinski.NumHDonors(m))


dp_ACHE = pd.DataFrame(descritors)
dp_ACHE.head()
```

	QED	MolWt	HeavyAtomCount	NumRotatableBonds	TPSA	NumHAcceptors	Num
0	0.795789	243.350	18	2	20.31	2	
1	0.834365	271.404	20	4	20.31	2	
2	0.422541	208.281	15	2	26.30	2	
3	0.603594	405.542	30	6	41.74	5	
4	0.805692	236.267	17	3	44.76	4	

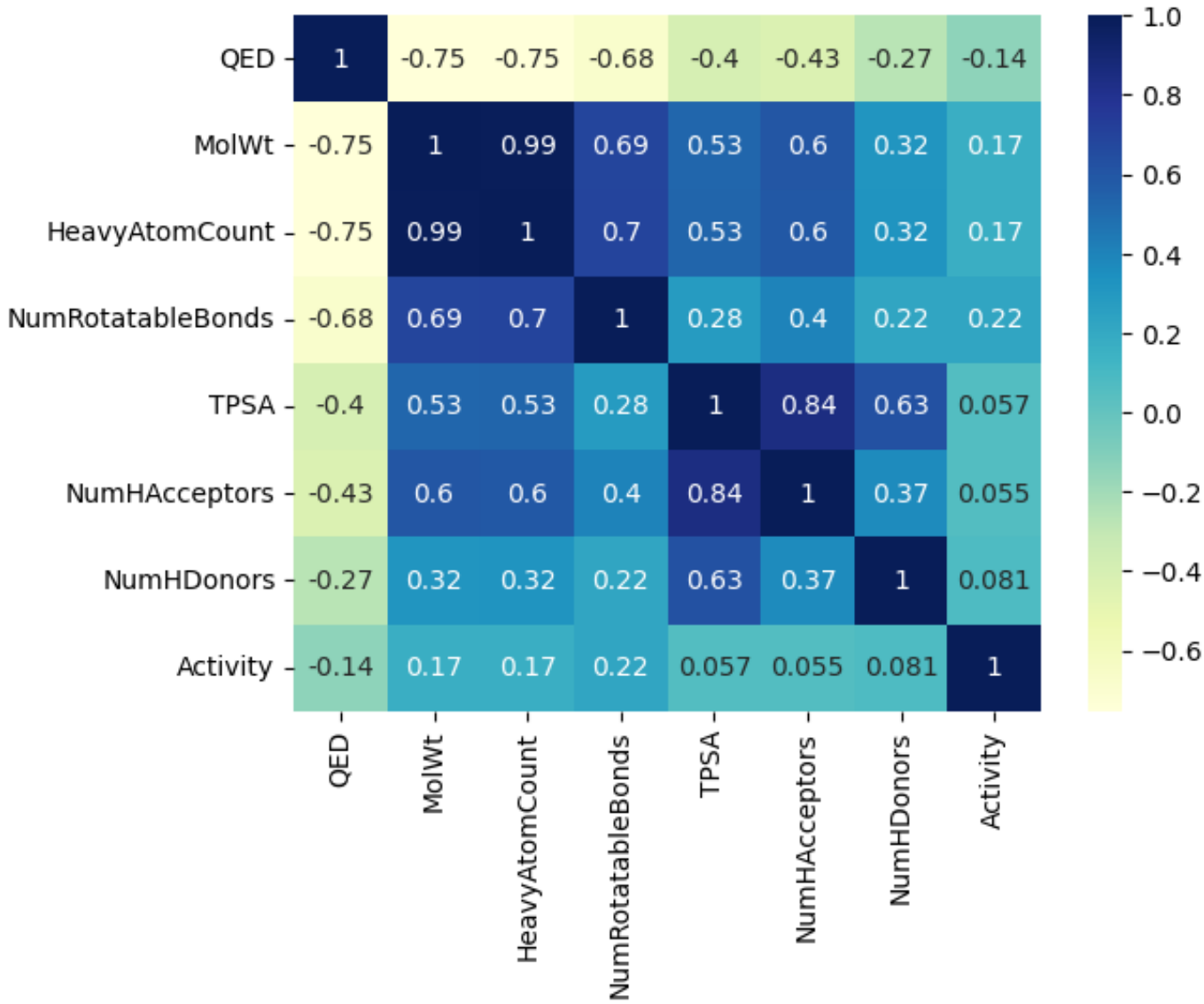
Next steps: Generate code with dp_ACHE | View recommended plots | New interactive sheet

```
dp_ACHE= dp_ACHE.assign(Activity=resultado['Activity'])
dp_ACHE.head()
```

	QED	MolWt	HeavyAtomCount	NumRotatableBonds	TPSA	NumHAcceptors	Num
0	0.795789	243.350	18	2	20.31	2	
1	0.834365	271.404	20	4	20.31	2	
2	0.422541	208.281	15	2	26.30	2	
3	0.603594	405.542	30	6	41.74	5	
4	0.805692	236.267	17	3	44.76	4	

Next steps: Generate code with dp_ACHE | View recommended plots | New interactive sheet

```
# plotting correlation heatmap
dataplot = sns.heatmap(dp_ACHE.corr(), cmap="YlGnBu", annot=True)

# displaying heatmap
plt.show()
```



```
# Nuevamente, igual que hicimos en clase vemos que hay una alta correlacion
# entre number of heavy atoms and the molecular mass asi que lo eliminamos.
dp_ACHE.drop(labels='HeavyAtomCount', axis=1, inplace=True)
dp_ACHE.head()
```

	QED	MolWt	NumRotatableBonds	TPSA	NumHAcceptors	NumHDonors	Activit
0	0.795789	243.350	2	20.31	2	0	1.
1	0.834365	271.404	4	20.31	2	0	1.
2	0.422541	208.281	2	26.30	2	0	0.
3	0.603594	405.542	6	41.74	5	0	1.
4	0.805692	236.267	3	44.76	4	0	1.

Next steps: Generate code with dp_ACHE | View recommended plots | New interactive sheet

Clasificacion o regresion

```
from sklearn.model_selection import train_test_split  # Para dividir dados em

# split data in 80:20 for train:test dataset
X = dp_ACHE.iloc[:, :-1]  # Todas as linhas, todas as colunas exceto a última
y = dp_ACHE.iloc[:, -1]   # Todas as linhas, apenas a última coluna (target)

# In the first step we will split the data in training and remaining dataset
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size=0.2, rando
```

```
print(X_train.shape), print(y_train.shape)
print(X_test.shape), print(y_test.shape)
```

```
(4306, 6)
(4306,)
(1077, 6)
(1077,)
(None, None)
```

```
# para evitar sesgos escalamos
from sklearn.preprocessing import *
sc = StandardScaler()

X_train = sc.fit_transform(X_train)


X_test = sc.transform(X_test)
print(X_test)
```

```
[[ 0.98264198 -0.18615117 -0.99321926  0.44433114  0.73334013 -0.10728556]
 [-0.55771635  0.40248466  0.28751606 -0.21938839  0.73334013 -1.00084772]
 [ 0.35103778  0.62094787 -0.77976338  0.18417339  0.23644594 -0.10728556]
 ...
 [ 1.80090835 -0.94036722 -0.3528516  -0.66673281  0.23644594 -0.10728556]
 [-0.37772239  0.1735376  -0.13939572 -0.47002817 -0.75734245 -1.00084772]
 [ 0.65127266 -1.3553322  -0.77976338 -1.55317275 -1.75113083 -1.00084772]]
```

```
# NO Usando Random Forest
# Usando Support Vector Machine
classifier = svm.SVC() #RandomForestClassifier(random_state=0)

classifier.fit(X_train, y_train)
```

```
▾ SVC
SVC()
```

Evaluacion

```
from sklearn import metrics
from sklearn.metrics import confusion_matrix, classification_report
def calc_metrics(labels_test, test_probs, threshold = 0.5):
    scores = [1 if x>=threshold else 0 for x in test_probs]
    auc = metrics.roc_auc_score(labels_test, test_probs, multi_class="ovr")
    kappa = metrics.cohen_kappa_score(labels_test,scores)
    confusion = metrics.confusion_matrix(labels_test,scores, labels=list(set(
    print('thresh: %.2f, kappa: %.3f, AUC test-set: %.3f'%(threshold, kappa, 
    print(confusion)
    print(metrics.classification_report(labels_test,scores,digits=3))
    return
```

```
x_pred = classifier.predict(X_test)
```

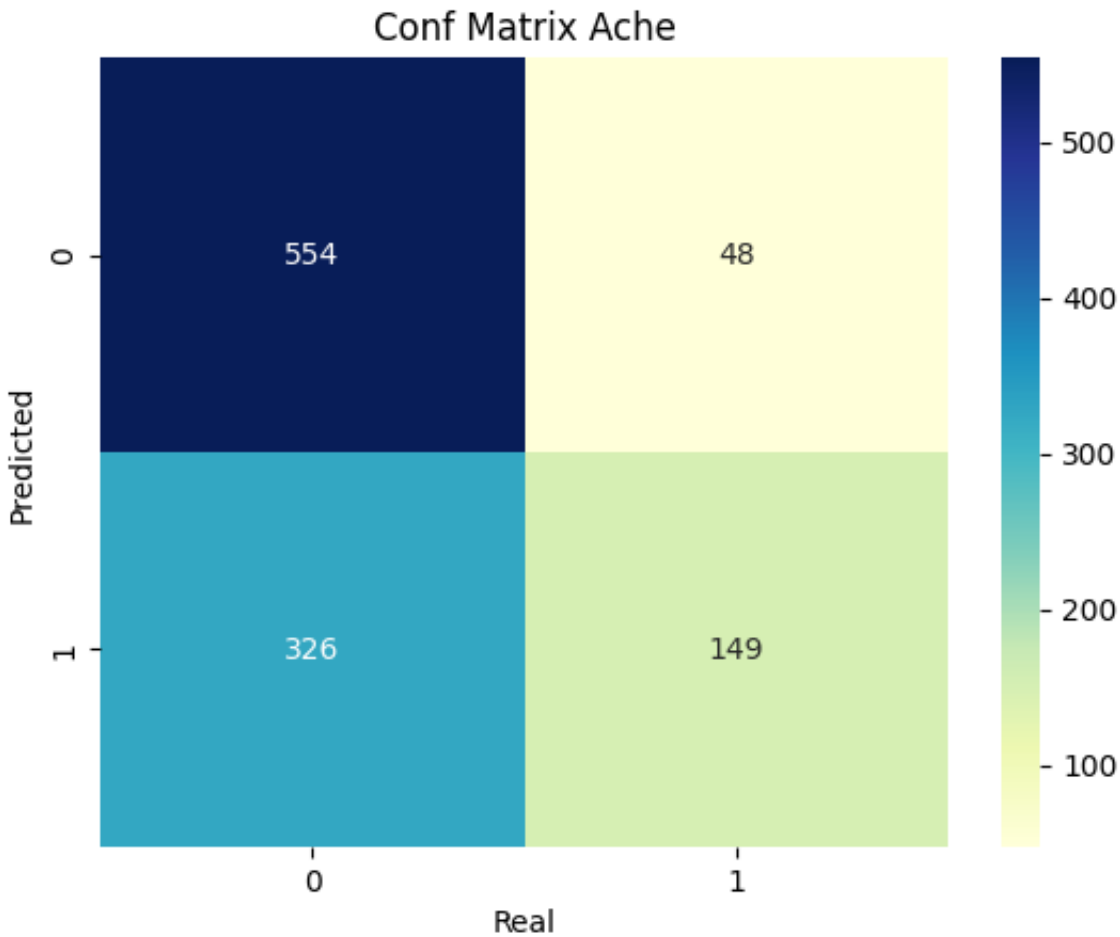
```
pd.DataFrame(x_pred).value_counts()
```

0	count
0.0	880
1.0	197

dtype: int64

```
print (calc_metrics(y_test, x_pred))

sns.heatmap(confusion_matrix(y_test, x_pred),cmap='YlGnBu' ,annot=True,fmt='2
plt.title('Conf Matrix Ache')
plt.ylabel('Predicted')
plt.xlabel('Real')
plt.show()
```

```
thresh: 0.50, kappa: 0.249, AUC test-set: 0.617
[[554  48]
 [326 149]]
              precision    recall  f1-score   support

         0.0      0.630     0.920     0.748       602
         1.0      0.756     0.314     0.443       475

    accuracy                          0.653      1077
   macro avg      0.693     0.617     0.596      1077
weighted avg      0.685     0.653     0.613      1077

None
```



fingerprints

```
# Notamos que no esta funcionando realmente asi que probaremos con fingerprin
# de las notas de clase
def to_numpyarray_to_list(desc):
    arr = np.zeros((1,))
    DataStructs.ConvertToNumpyArray(desc, arr)
    return arr.tolist()

def calcula_fp1(dataframe):
    d={}
    c=0
    for x in dataframe:

        mol = Chem.MolFromSmiles(x)
        fp1 = rdMolDescriptors.GetMorganFingerprintAsBitVect(mol, 20, nBits=2
        FP_as_BitVect = to_numpyarray_to_list(fp1)
        d['result{}'.format(c)] = FP_as_BitVect
        c+=1
    df = pd.DataFrame(d).transpose()
    return df


fp_ACHE = calcula_fp1(smiles_ache_stand)
fp_ACHE
```

Streaming output truncated to the last 5000 lines.

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result5378	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0
result5379	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0
result5380	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	1.0
result5381	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0
result5382	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0

5383 rows × 2048 columns

```
# Resetar índices antes de atribuir
fp_ACHE = fp_ACHE.reset_index(drop=True)
resultado = resultado.reset_index(drop=True)
fp_ACHE['Activity'] = resultado['Activity']
fp_ACHE.head()
```

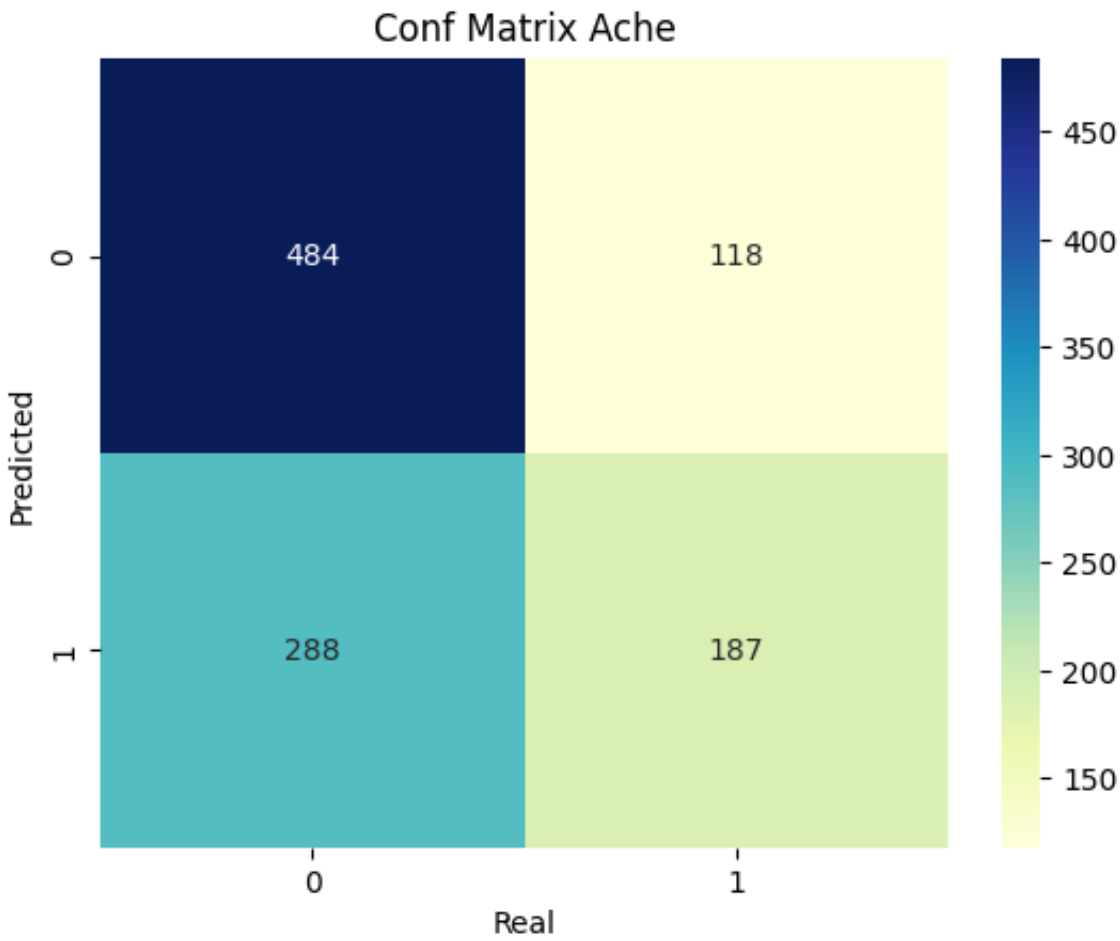
	0	1	2	3	4	5	6	7	8	9	...	2039	2040	2041	2042	2043	2(
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	

5 rows × 2049 columns

```
# Usando NaiveBayes
classifier = GaussianNB()

classifier.fit(X_train, y_train)
x_pred = classifier.predict(X_test)


print (calc_metrics(y_test, x_pred))

sns.heatmap(confusion_matrix(y_test, x_pred),cmap='YlGnBu' ,annot=True,fmt='2
plt.title('Conf Matrix Ache')
plt.ylabel('Predicted')
plt.xlabel('Real')
```

```
plt.show()
```

```
thresh: 0.50, kappa: 0.205, AUC test-set: 0.599
[[484 118]
 [288 187]]
              precision    recall  f1-score   support

         0.0      0.627     0.804     0.705       602
         1.0      0.613     0.394     0.479       475

    accuracy                          0.623      1077
   macro avg      0.620     0.599     0.592      1077
weighted avg      0.621     0.623     0.605      1077

None
```



Usando Chemberta

```
from transformers import AutoTokenizer, AutoModelForMaskedLM
```

```
import torch
from tqdm import tqdm  # Para barra de progresso


# 1. Carregar modelo e tokenizador
tokenizer = AutoTokenizer.from_pretrained("DeepChem/ChemBERTa-77M-MLM")
model = AutoModelForMaskedLM.from_pretrained("DeepChem/ChemBERTa-77M-MLM")
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

# 2. Função para gerar embeddings
def get_molecule_embedding(smiles, model, tokenizer):
    inputs = tokenizer(smiles, return_tensors="pt", padding=True, truncation=
    with torch.no_grad():
        outputs = model(**inputs, output_hidden_states=True)
        last_hidden_state = outputs.hidden_states[-1]
        #return last_hidden_state[:,0,:].squeeze().cpu().numpy() #get the fir
        return last_hidden_state.mean(dim=1).squeeze().cpu().numpy() #average

# 3. Processar todos os SMILES (com barra de progresso)
embeddings = []
failed_smiles = []

for smile in tqdm(smiles_ache_stand, desc="Gerando embeddings"):
    try:
        if pd.isna(smile) or not smile:  # Pula valores nulos/vazios
            continue
        emb = get_molecule_embedding(smile, model, tokenizer)
        embeddings.append(emb)
    except Exception as e:
        failed_smiles.append(smile)
        continue

# 4. Converter para DataFrame
df_embeddings = pd.DataFrame(embeddings)
#df_embeddings['SMILES'] = [s for s in smiles_ache_stand if s not in failed_s
df_embeddings['Activity'] = resultado['Activity']
print(f"\nEmbeddings gerados: {len(embeddings)}")
print(f"SMILES falhados: {len(failed_smiles)}")
print(f"Shape dos embeddings: {df_embeddings.shape}")
```

```
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: Use
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access pu
  warnings.warn(
tokenizer_config.json: 100%                                    1.27k/1.27k [00:00<00:00, 31.6kB/s]
config.json: 100%                                              631/631 [00:00<00:00, 10.6kB/s]
vocab.json: 100%                                               6.96k/6.96k [00:00<00:00, 166kB/s]
merges.txt: 100%                                               52.0/52.0 [00:00<00:00, 2.47kB/s]
tokenizer.json: 100%                                           8.26k/8.26k [00:00<00:00, 195kB/s]
added_tokens.json: 100%                                        25.0/25.0 [00:00<00:00, 412B/s]
special_tokens_map.json: 100%                                  420/420 [00:00<00:00, 12.7kB/s]
pytorch_model.bin: 100%                                        13.7M/13.7M [00:00<00:00, 62.3MB/s]
Gerando embeddings:   1%|          | 49/5383 [00:01<02:15, 39.50it/s]
model.safetensors: 100%                                        13.7M/13.7M [00:00<00:00, 36.2MB/s]
Gerando embeddings: 100%|██████████| 5383/5383 [01:53<00:00, 47.58it/s]

Embeddings gerados: 5383
SMILES falhados: 0
Shape dos embeddings: (5383, 385)
```

```
df_embeddings.head()
```

	0	1	2	3	4	5	6	7	
0	-0.038857	0.003095	-0.007130	0.051530	0.144723	-0.024886	0.061918	0.090813	0.0
1	-0.061092	-0.047996	-0.033261	0.057001	0.127676	-0.018489	0.029460	0.039060	-0.0
2	0.024234	-0.018762	-0.075961	0.140852	0.069428	-0.020260	0.045222	0.195928	-0.0
3	0.053842	-0.134951	-0.011490	0.017317	0.077888	0.009328	0.064008	0.204522	-0.0
4	-0.114140	-0.012006	-0.030179	0.057491	-0.079374	0.064951	0.001323	0.068954	0.0

5 rows × 385 columns

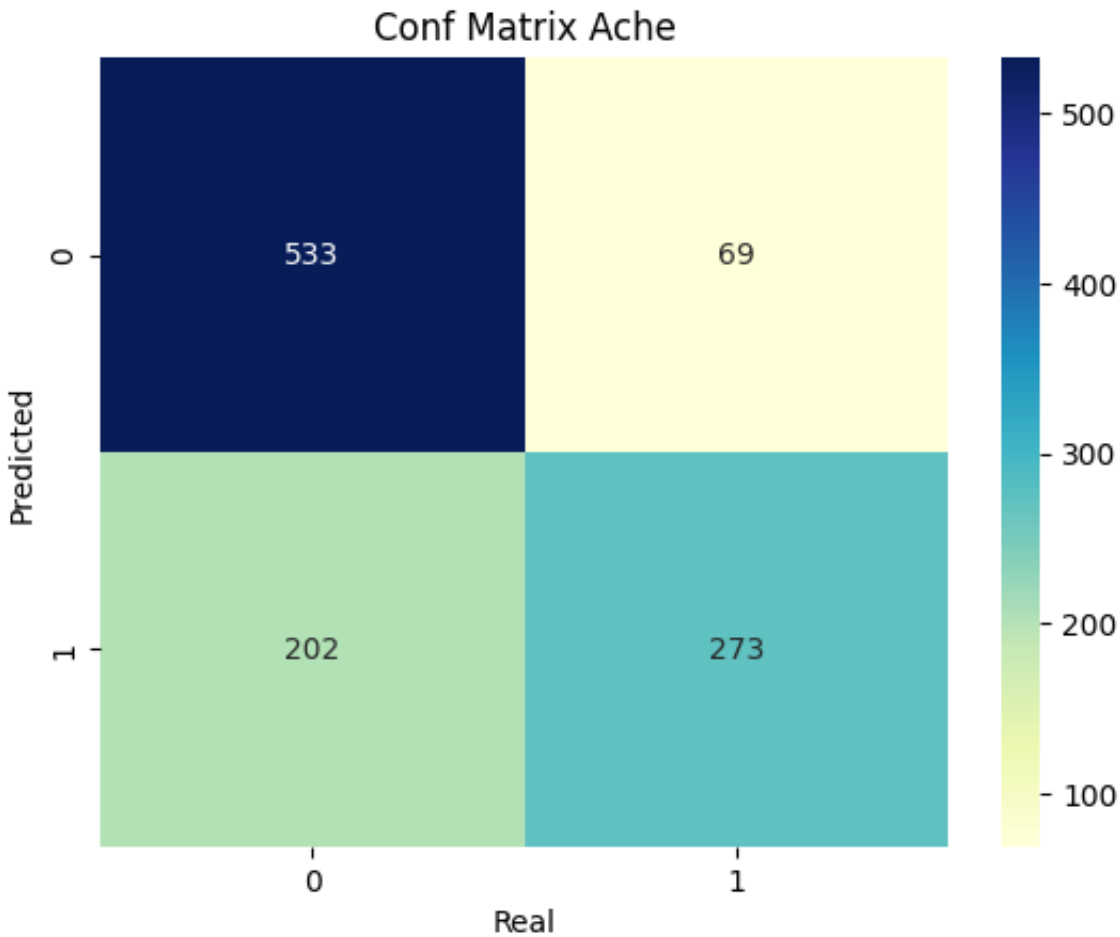
```
# split data in 80:20 for train:test dataset
X = df_embeddings.iloc[:, :-1]  # Todas as linhas, todas as colunas exceto a
y = df_embeddings.iloc[:, -1]   # Todas as linhas, apenas a última coluna (ta

# In the first step we will split the data in training and remaining dataset
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size=0.2, rando

print(X_train.shape), print(y_train.shape)
print(X_test.shape), print(y_test.shape)
```

```
(4306, 384)
(4306,)
(1077, 384)
(1077,)
(None, None)
```

```
# Probando nuevamente con SVM
classifier = svm.SVC()

classifier.fit(X_train, y_train)
x_pred = classifier.predict(X_test)
```

```
print (calc_metrics(y_test, x_pred))

sns.heatmap(confusion_matrix(y_test, x_pred),cmap='YlGnBu' ,annot=True,fmt='
plt.title('Conf Matrix Ache')
plt.ylabel('Predicted')
plt.xlabel('Real')
plt.show()
```

```
thresh: 0.50, kappa: 0.474, AUC test-set: 0.730
[[533  69]
 [202 273]]
              precision    recall  f1-score   support

         0.0      0.725     0.885     0.797       602
         1.0      0.798     0.575     0.668       475

    accuracy                          0.748      1077
   macro avg      0.762     0.730     0.733      1077
weighted avg      0.757     0.748     0.740      1077

None
```




```
# Observamos que naive bayes funcono bastante bien con fingerprints
# en comparacion con propiedades moleculares con SVM
#  y/o chemberta independientemente del clasificador.
```

Start coding or generate with AI.